

AYE  
TECH HUB

• SOLAR ENERGY • LESSON 08

# DC and AC Solar Systems

Off-Grid · On-Grid · Hybrid · DC Coupling · AC Coupling

Lesson 08 of 30 | Solar Energy Program

**Awet G. Nway**

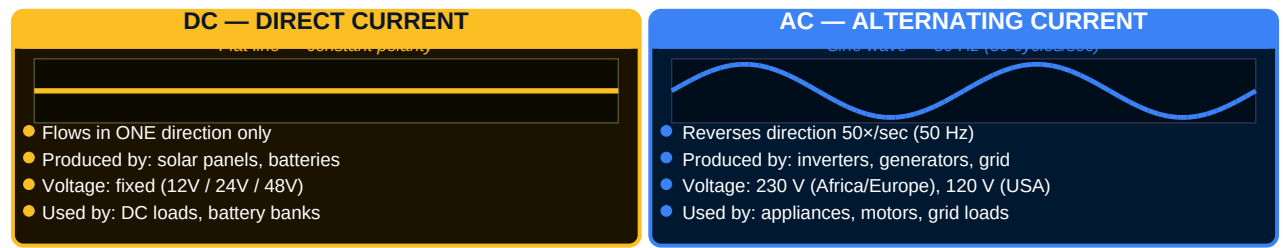
Founder, AYE Tech Hub | Solar Energy & Electrical Engineering Specialist

- DC vs AC electricity: fundamental difference and why solar uses both
- DC system architecture: panels → charge controller → battery → load
- AC system architecture: panels → inverter → grid or AC loads
- Off-grid, on-grid, and hybrid — three solar system topologies compared
- DC coupling vs AC coupling: which wiring approach and when to use each

# DC and AC Electricity in a Solar System

Solar panels produce DC electricity. The grid runs on AC. Batteries store DC. Most appliances in your home run on AC. **A solar system is therefore a DC/AC conversion system — managing the flow between these two electrical currencies.** Understanding DC and AC is the foundation for understanding every solar wiring diagram you will ever encounter.

Figure 1 — DC vs AC Electricity: The Two Electrical Currencies of a Solar System



| Property                | DC (Direct Current)                                                                                           | AC (Alternating Current)                                                                                                      |
|-------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Direction of flow       | Always the same direction — from positive (+) to negative (–) terminal                                        | Reverses direction 50 times per second (50 Hz in Africa and Europe)                                                           |
| Waveform                | Flat constant line — voltage stays the same polarity at all times                                             | Sine wave — voltage swings from +325V peak to –325V peak at 230V RMS                                                          |
| Voltage levels (solar)  | 12V, 24V, 48V system voltages (battery nominal). Panels: 20–60V open circuit per panel.                       | 230V single-phase (Africa, Europe, Asia). 120V/240V split-phase (USA). 400V three-phase (commercial).                         |
| Produced by             | Solar panels (photovoltaic effect), batteries (chemical energy), DC generators, rectifiers (AC→DC conversion) | Inverters (DC→AC conversion), alternators, transformers (voltage stepping), the national grid                                 |
| Transmission efficiency | Loses voltage over long distances due to resistance. Practical for short distances only (within a property).  | Can be stepped up to high voltage (11kV, 33kV, 400kV) for long-distance transmission with minimal loss — why the grid uses AC |
| Used by in solar        | Charging batteries, DC-direct loads (LED lights, DC fans, pumps with DC motors, USB chargers)                 | All standard household appliances: refrigerators, TVs, ACs, washing machines, computers via AC adaptor                        |

**Why 48V systems are preferred for larger solar installations:  $Power = Voltage \times Current$  ( $P = V \times I$ ). At 48V, the same power requires half the current compared to 24V. Half the current means quarter the cable losses ( $losses = I^2 \times R$ ). This allows thinner, cheaper cables and reduces heat — which is why all serious off-grid and hybrid systems above 3 kW use 48V.**

# Three Solar System Topologies

There are three fundamental ways to design a solar power system, each with a different relationship between panels, batteries, inverter, and grid. Choosing the right topology depends on whether grid power is available, how reliable the grid is, and the budget.

Figure 2 — Three Solar System Topologies: Off-Grid vs On-Grid vs Hybrid



| Feature                               | Off-Grid System                                 | On-Grid (Grid-Tie)                                | Hybrid System                                         |
|---------------------------------------|-------------------------------------------------|---------------------------------------------------|-------------------------------------------------------|
| Grid connection                       | None — standalone                               | Required — no battery                             | Optional grid backup                                  |
| Battery bank                          | Essential — main backup                         | None (standard)                                   | Required                                              |
| Works at night                        | Yes — from battery                              | No (no battery)                                   | Yes — from battery                                    |
| Works during grid outage              | Always (no grid dependency)                     | No — shuts off (safety)                           | Yes — switches to battery                             |
| Can sell power to grid (net metering) | No                                              | Yes — main benefit                                | Yes — sells excess after battery full                 |
| Inverter type needed                  | Off-grid / standalone inverter-charger          | Grid-tie inverter (anti-islanding)                | Hybrid / multi-mode inverter-charger                  |
| Best suited for                       | Remote areas, rural communities, no grid access | Urban buildings with reliable grid — reduce bills | Areas with unreliable grid, businesses needing backup |
| Typical cost vs on-grid               | 40–60% higher (battery bank adds cost)          | Lowest upfront cost                               | 30–50% more than on-grid                              |

**Most common in East Africa and the Horn of Africa: Hybrid systems — because grid power exists in cities but is unreliable (frequent outages lasting 4–8 hours daily). A hybrid system with a 10–20 kWh battery bank provides power during outages while also reducing electricity bills during grid hours.**

## DC Coupling vs AC Coupling

Within a hybrid or off-grid system, there are two ways to connect the solar panels to the battery bank — DC coupling and AC coupling. This decision affects efficiency, expandability, and cost. Most new installations choose DC coupling; AC coupling is common when retrofitting batteries onto an existing grid-tie system.

Figure 3 — DC Coupling vs AC Coupling: Wiring Architecture Comparison



| Factor                | DC Coupling                                                                          | AC Coupling                                                                                 |
|-----------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Charging path         | Panels → MPPT Charge Controller → Battery (DC throughout)                            | Panels → Grid-tie inverter → AC bus → Battery inverter → Battery (DC→AC→DC)                 |
| Efficiency            | 93–97% — one DC→DC conversion via MPPT. Fewer conversion steps.                      | 85–92% — two inversions (DC→AC, then AC→DC). Some energy lost.                              |
| Best for              | New installations: house not yet wired. Simple, efficient, industry standard choice. | Retrofitting: existing grid-tie system. Adding batteries to a working on-grid installation. |
| Expandability         | Limited by charge controller string voltage (typically 150V or 450V max)             | Easier to add more panels without changing battery-side components                          |
| Cost                  | Slightly less equipment for new builds — one fewer inverter                          | Requires two separate inverters (grid-tie + battery inverter) — higher cost                 |
| Generator integration | Generator charges battery via battery inverter AC input — simple                     | Generator integrates into AC bus easily — flexible                                          |

| Component                  | DC System Role                                                                                         | AC System Role                                                                                     | Key Specification                                                                     |
|----------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Solar Panels               | Source of DC power — produce 20–60V per panel                                                          | Same — source of DC power                                                                          | Voc (open circuit voltage), Isc (short circuit current), Pmax                         |
| MPPT Charge Controller     | DC coupling only: steps down panel voltage to battery voltage efficiently                              | Not needed on panel side — grid-tie inverter handles MPPT                                          | Must match string Voc; rated in amps (battery output current)                         |
| Grid-Tie / Hybrid Inverter | Hybrid inverter: converts battery DC → AC for loads, manages MPPT, charges battery from grid/generator | Grid-tie inverter: converts panel DC → AC for grid. Battery inverter: charges battery from AC bus. | VA rating ≥ peak load. Must have grid-tie certification (IEC 62109)                   |
| Battery Bank               | Stores DC energy. Connected to charge controller output.                                               | Stores DC energy. Connected to battery inverter output.                                            | Capacity in kWh = Ah × Voltage ÷ 1000. Depth of discharge determines usable capacity. |
| AC Distribution Board      | Receives AC from inverter output — distributes to household circuits                                   | Central AC bus connects grid-tie inverter, battery inverter, grid, and loads                       | MCBs (circuit breakers) sized per circuit. Main breaker ≥ inverter output current.    |

**What comes next — SOLAR-009: Solar Wiring Basics — how to wire solar panels in series, parallel, and series-parallel configurations; string sizing, fuse and cable sizing for DC wiring, grounding, surge protection, and safe DC disconnect switches. The practical wiring knowledge that turns system design into a real installation.**